

A Ablation Studies and Additional Experiments

A.1 Continuation Evaluation

Here we extend the continuation evaluation in Fig. 1 to include two additional instruct-model-based controls: *Instruct + HIP* and *Instruct + FT*. The setup is otherwise the same as in Section 4: we prompt each model with the same human and AI prefixes and score only the generated continuation. *Instruct + HIP* denotes an instruction-tuned model after HIP adaptation. In this experiment, however, we do not use that checkpoint as a paraphraser. Instead, we use it simply as a generator, exactly as we do for the raw base and raw instruct models. *Instruct + FT* denotes a separate control in which we further fine-tune the same instruct model on FineWeb-Edu [Penedo et al., 2024] with a plain language-modeling objective rather than the HIP paraphrase objective.

As Fig. 6 shows, the same qualitative pattern holds in both model families and under both human and AI prefixes: instruction tuning makes continuations look much less human to GPTZero and Pangram, while both *Instruct + HIP* and *Instruct + FT* recover a large portion of that lost human-likeness. This suggests that detector judgments are sensitive to post-training state itself, not only to the explicit paraphrase pipeline. At the same time, *Instruct + FT* does not produce a usable paraphraser, so recovering human-like continuation behavior alone is not sufficient for the main HIP objective.



Figure 6: GPTZero and Pangram human-probability scores on text generated by Llama3-8B and Qwen3-8B variants when conditioned on human-written or AI-generated prefixes. The error bars show 95% confidence intervals. In both families, the base model is judged much more human than the instruct model, and both *Instruct + HIP* and *Instruct + FT* recover much of that lost human-likeness.

A.2 HIP on OpenAI Models via the Fine-Tuning API

We also test whether HIP transfers to a closed-weight model fine-tuned through the OpenAI API. Starting from GPT-4.1-nano, we fine-tune the model on the same paired paraphrase dataset $\mathcal{D}$ used in the main experiments, but now through the OpenAI fine-tuning API rather than local LoRA-based training. We then run the same 10-round iterative paraphrasing evaluation on the same 256-example evaluation set and score the outputs with the same semantic, GPTZero, and Pangram metrics.

As Fig. 7 shows, HIP does not produce a useful semantic-evasion trade-off in this setting. Semantic scores remain high throughout the 10 rounds, but detector-assigned human probability quickly falls and remains low on both GPTZero and Pangram. In other words, the model continues to paraphrase coherently, but the iterative pipeline does not move its outputs toward the human-like region that HIP reaches on open-weight Llama and Qwen models in Section 4.

Combined with the main results and the continuation evaluation, one plausible hypothesis is that fine-tuning through the OpenAI API does not induce the same low-distortion base-model behavior that HIP exploits in open-weight models. A likely reason is that the platform may mix in additional post-training or alignment-related data or procedures during training and serving. We cannot verify that mechanism directly, but the negative result is consistent with that interpretation.

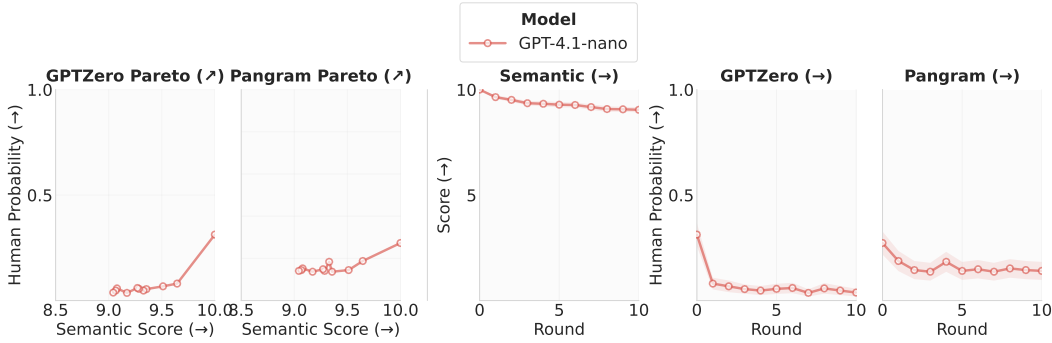


Figure 7: HIP on OpenAI GPT-4.1-nano by running through the OpenAI fine-tuning API. The first two subplots show the GPTZero and Pangram Pareto frontiers for the trade-off between semantic preservation and human-likeness, while the last three show semantic score, GPTZero human probability, and Pangram human probability over the rounds, respectively. Shaded areas indicate 95% confidence intervals. Unlike the open-weight Llama and Qwen models in the main paper, the trade-off remains weak under both detectors despite relatively high semantic preservation.

A.3 HIP on Instruct Models with Native Chat Templates

We also test whether the humanizing effect of HIP depends on the simplified source-target formatting used in Section 3.2 rather than on an instruct model’s native chat interface. Starting from the Llama3-8B-Instruct and Qwen3-8B-Instruct checkpoints, we keep the same paired paraphrase dataset $\mathcal{D}$ and 10-round evaluation, but express both training and inference through each model’s native chat template. Concretely, the system prompt instructs the model to paraphrase naturally while preserving meaning, and the user message contains the passage to be rewritten. We then evaluate the resulting chat-template paraphraser on the same 256-example evaluation set with the same metrics.

As Fig. 8 shows, this change has only a modest effect on the trade-off. In both cases, the overall Pareto frontier remains close to that of standard HIP, and the qualitative behavior is unchanged.

Taken together, these results suggest that the main HIP effect does not depend strongly on whether instruct models are adapted and run through plain source-target formatting or through their native chat templates. The non-chat formulation used in Section 3.2 is not necessary for instruct models.

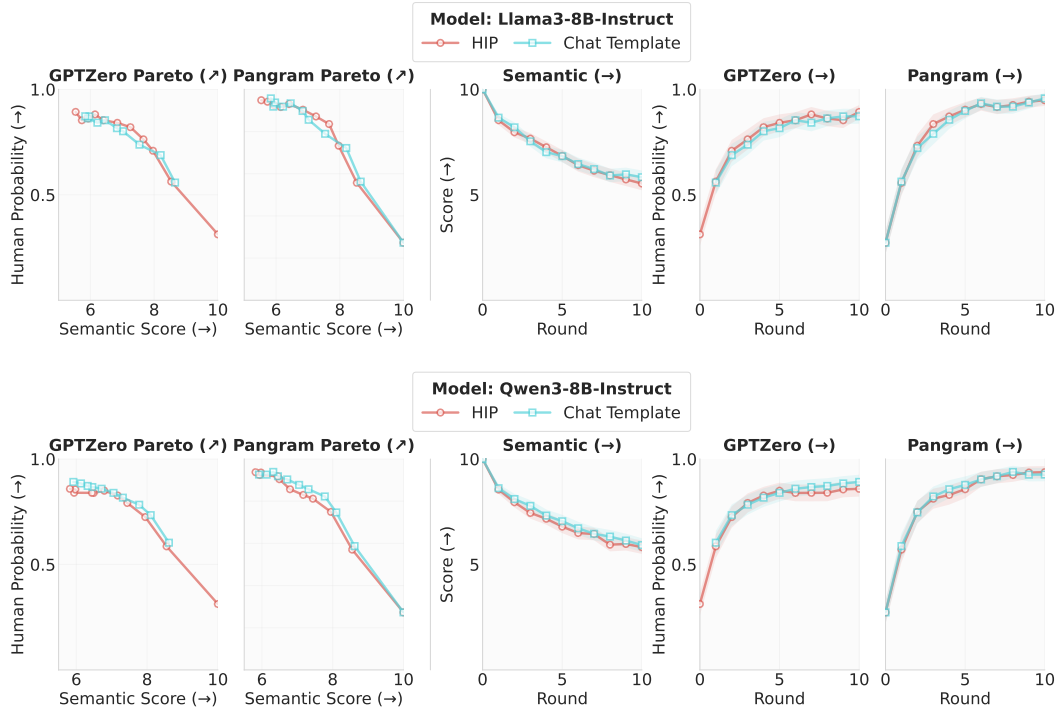


Figure 8: Standard HIP versus the native-chat-template variant on Llama3-8B-Instruct and Qwen3-8B-Instruct. In each panel, the first two subplots show the GPTZero and Pangram Pareto frontiers for the trade-off between semantic preservation and human-likeness, while the last three show semantic score, GPTZero human probability, and Pangram human probability over the rounds, respectively. Shaded areas indicate 95% confidence intervals. The differences are modest in both families, suggesting that the main HIP effect is not driven by the absence of a chat template when training an instruct model.